

	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		ASSIGNMENT	
Branch	Computer Science and Engineering,	Programme	B.Tech	
Course Name	Semiconductor Devices and Fabrication Theory	Semester	VIII	
Course Code	OPL4006	Year/Period	2026/Even	
ASSIGNMENT-02	Submit All Assignments	Maximum Marks	10	GP-6
Learning Level (LL)	L1: Remembering	L3: Application	L5: Evaluation	
	L2: Understanding	L4: Analysis	L6: Creation	
Q. No.	Questions		COs	LL
1	Enumerate the drawbacks of vacuum tube devices and the need for integrated circuits (ICs).		CO4	L4
2	Trace the evolution of the integrated circuit and explain the stages of integration, in terms of device density and dimensions.		CO5	L2
3	Explain the meaning of Moore's law and its dominance in the semiconductor industry.		CO5	L3
4	List the various states in integrated circuit manufacturing.		CO5	L3
5	Mention 4 challenges in IC manufacturing.		CO5	L4
6	Explain the conversion of silicon oxide into poly crystalline silicon.		CO5	L2
7	Distinguish between metallurgical grade and electronic grade silicon and list the steps for purification of metallurgical grade silicon.		CO5	L2
8	Enumerate the steps for converting single crystal ingot into silicon wafers.		CO5	L4
9	Distinguish between Czochralski and float zone technique processes. Why do you think the float zone technique is harder to implement for larger wafers?		CO5	L3
10	Classify fabrication operations into four main categories and define terms such as die, chip, and scribe lines.		CO6	L3
11	Explain the types of layering operations and enumerate the various layers found in an integrated circuit.		CO6	L3
12	Explain the basics of photolithography and the difference between positive and negative resists and how they are used in patterning.		CO6	L4
13	Differentiate between the two main doping techniques in terms of their advantages and limitations.		CO6	L2
14	Enumerate the various steps in the fabrication of a metal oxide semiconductor field-effect transistor starting from a silicon wafer and classify the steps into the four categories.		CO6	L5
15	What is Oxidation? Describe briefly on Oxidation processes.		CO6	L4
16	Explain Thermal Diffusion with neat diagram?		CO6	L4
17	Explain briefly on Ion implantation?		CO6	L3

18	State the difference between various doping techniques.	CO6	L4
19	Explain the stages of the lithography process with suitable diagrams.	CO6	L2
20	What do you mean by a mask? Explain the steps involved in mask making using the flow diagram.	CO6	L3
21	Write the components of a photoresist. What do you mean by positive and negative photoresists?	CO6	L3
22	How does conventional lithography differ from E-beam lithography?	CO6	L5
23	Explain the challenges of Wet Etching. According to the applicability, what are the limitations of Wet Etching?	CO6	L2
24	Explain the process of Dry Etching with the help of a diagram.	CO6	L2
25	How many layers exist in the Deposition process?	CO6	L3
26	Write down some important film parameters helps in the growth technique.	CO6	L2
27	Distinguish between CVD and ALD. What are the advantages of ALD over CVD and its main limitation?	CO6	L3
28	Briefly explain the Copper Dual-Damascene Process.	CO6	L2
29	List the evolution of metallization materials from the earliest ICs to present. Explain the reasons for the change in each case.	CO6	L2
30	Enumerate the significance of FEOL and BEOL with reference to IC fabrication.	CO6	L5

Assignment -02	Topic: Transistors, Development of ICs , Silicon Wafer Manufacturing , Integrated Circuit Fabrication , Oxidation & Doping , Lithography , Etching and Deposition , Metallization & Polishing	Date of Assignment - 02: 25.04.26	Date of submission: 05.05.26
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Note:

1. The Assignments in total carry weightage of **20 marks out of 100**
2. Course outcome CO5 to CO6 was covered.

Course Outcomes	CO1	Explain the electron statistics, semiconductor material properties to underlying physics.
	CO2	Compare and analyze charge carrier dynamics in an intrinsic as well as extrinsic Semiconductor and Fundamentals of Semiconductors.
	CO3	Describe the metal-semiconductor junction to charge carrier action and conductivity.
	CO4	Analyze and compare the characteristics of various electronic devices (Diode, BJT, JFET, MOSFET)
	CO5	Justify the techniques of development of ICs emphasizing the silicon wafer manufacturing as well as fabrication of integrated circuits.
	CO6	Distinguish various steps behind fabrication of ICs (oxidation, lithography as well as etching and deposition) and outline the challenges.